

BRIEF REPORT

Prevalence of High-Risk CTA-Based Carotid Plaque-RADS Subtypes in Patients With Embolic Stroke of Undetermined Source

Jae W. Song¹ MD, MS; Huy Q. Phi¹ BS; Manisha Koneru¹ MD, MS; Quy Cao¹ PhD; Jeremy Rubin¹ MS; Yu Sakai¹ MD; Lamya Ibrahim¹ MD; Sonya E. Zhou¹ MD; John H. Woo¹ MD; Scott E. Kasner¹ MD; Luca Saba¹ MD; Brett L. Cucchiara¹ MD

BACKGROUND: A modified computed tomography angiography (CTA)-based Carotid Plaque Reporting and Data System (Plaque-RADS) classification was applied to a cohort of patients with embolic stroke of undetermined source to test whether high-risk Plaque-RADS subtypes are more prevalent on the ipsilateral side of stroke. With the widespread use of CTA for stroke evaluation, a CTA-based Plaque-RADS would be valuable for generalizability.

METHODS: A retrospective observational cross-sectional study was conducted at a single integrated health system comprised of 3 hospitals with a comprehensive stroke center between October 1, 2015, and April 1, 2017. Patients with unilateral anterior circulation stroke and <50% carotid stenosis on CTA were retrospectively identified. Maximum plaque thickness and ulceration were assessed by a neuroradiologist blinded to the stroke side. A semiautomated segmentation software measured intraplaque hemorrhage volumes. Modified CTA-based Plaque-RADS classification was defined as (1) no plaque, (2) plaque thickness <3 mm, (3) plaque thickness ≥3 mm or ulcerated, and (4) plaque with intraplaque hemorrhage >50 mm³ irrespective of plaque thickness. High-risk plaque subtypes (Plaque-RADS 3 and 4) were compared with low-risk subtypes (Plaque-RADS 1 and 2).

RESULTS: Ninety-four patients (55% women; median age, 66 years) were included. CTA-based Plaque-RADS categories for plaques ipsilateral to the stroke side were as follows: (1) 14.9%, (2) 42.6%, (3) 41.5%, and (4) 1.1%. Carotid plaques contralateral to stroke side were Plaque-RADS: (1) 21.3%, (2) 46.8%, (3) 31.9%, and (4) 0%. When compared with the contralateral side, plaques ipsilateral to the stroke side were significantly associated with high-risk Plaque-RADS subtypes in a mixed-effects logistic model adjusting for age and sex (adjusted odds ratio, 2.10 [95% CI, 1.20–3.71]; $P=0.01$).

CONCLUSIONS: Carotid plaque ipsilateral to the stroke side was significantly associated with CTA-based high-risk Plaque-RADS subtypes in an embolic stroke of undetermined source cohort. A CTA-based Plaque-RADS classification may be useful for identifying potentially causative carotid plaque phenotypes in patients with embolic stroke of undetermined source.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: atherosclerosis ■ carotid stenosis ■ embolic stroke ■ hemorrhage ■ hospitals ■ stroke

The Carotid Plaque Reporting and Data System (Plaque-RADS) classification is a novel stroke risk scoring system recently introduced to standardize communication in terminology and information among clinicians and researchers.¹ It stratifies plaques into 4

subtypes, incorporating plaque composition and imaging features using a multimodal imaging approach (ultrasound, computed tomography [CT], and magnetic resonance [MR]). It could be useful in clinical practice for risk stratification and identification of potentially causative plaque

Correspondence to: Jae W. Song, MD, MS, Department of Radiology, University of Pennsylvania, 3400 Spruce St, Philadelphia, PA 19104. Email jae.song@pennmedicine.upenn.edu

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Nonstandard Abbreviations and Acronyms

CT	computed tomography
CTA	computed tomography angiography
ESUS	embolic stroke of undetermined source
IPH	intraplaque hemorrhage
MPT	maximum plaque thickness
MR	magnetic resonance
PARISK	Plaque at RISK
Plaque-RADS	Plaque Reporting and Data System

phenotypes.¹ However, Plaque-RADS 4 subclassifications require the identification of intraplaque hemorrhage (IPH; 4A), ruptured fibrous cap (4B), and intraluminal thrombus (4C), features best identified by MR vessel wall imaging, which has limited availability. CT angiography (CTA) is widely used for stroke assessment, and thus a CTA-based Plaque-RADS may have broader utility. We developed a modified CTA-based Carotid Plaque-RADS system and applied it to a cohort of patients with embolic stroke of undetermined source (ESUS).

METHODS

The data that support the findings of this study are available from the corresponding author upon reasonable request. We used the STROBE guidelines (Strengthening the Reporting of Observational Studies in Epidemiology) to report this study.² This study was approved by our local institutional review board. A retrospective observational cohort of consecutive patients with a diagnosis of acute ischemic stroke and admitted between October 1, 2015, and April 1, 2017, in 1 of 3 hospitals (Hospital of the University of Pennsylvania, Penn Presbyterian Medical Center, and Pennsylvania Hospital) within the Penn Medicine Health System was identified.³ Patients ≥ 18 years of age with unilateral anterior circulation ischemic stroke, imaged with neck CTA, and who met criteria for ESUS as determined by a vascular neurologist were included in this study.³ Exclusion criteria included acute infarction in >1 vascular territory, prior carotid revascularization, occlusion of either cervical internal carotid artery, or CTA performed >10 days after stroke onset. Demographic and clinical data were collected from electronic medical records.

CTA neck imaging was acquired in the axial plane using a fourth-generation helical CT scanner with a section thickness ranging from 0.625 to 1.5 mm, a matrix size of 512 $\times$ 512, and a field of view of 20 to 33 cm. Iodinated contrast (100 mL Isovue-370) was administered intravenously via a 20-gauge antecubital catheter. Blinded to the clinical data and stroke side, a neuroradiologist measured percent stenosis based on the North American Symptomatic Carotid Endarterectomy Trial criteria, maximum plaque thickness (MPT), and scored for plaque ulceration across a 4-cm segment of the carotid bifurcation. A semiautomated plaque quantification software (Elucid Bioimaging, Boston, MA) segmented IPH volumes and those were checked by a neuroradiologist.⁴

Plaque-RADS (Table S1) classifies subtypes as (1) normal vessel wall; (2) maximum wall thickness <3 mm; (3) maximum wall thickness >3 mm without complex features (3A), thin fibrous cap (3B), and ulceration (3C); and (4) IPH (4A), ruptured fibrous cap (4B), and intraluminal thrombus (4C). We modified the classification using CTA imaging features of MPT, ulcerations, and IPH volumes as follows: (1) normal vessel wall, (2) MPT <3 mm, (3) MPT ≥ 3 mm or ulcerated, and (4) IPH >50 mm³ irrespective of MPT. A 50-mm³ threshold was chosen based on a prior study showing this threshold had the strongest association with ipsilateral cerebrovascular symptoms.⁵ Plaque-RADS 3 and 4 are high-risk subtypes. The Figure shows examples of the modified Plaque-RADS classification.

Statistical Analysis

Continuous variables are reported as medians with interquartile ranges and categorical variables as frequencies. Differences in baseline demographics were compared across Plaque-RADS subtypes within ipsilateral and contralateral plaque substrata using the Kruskal-Wallis or χ^2 test.

A mixed-effects logistic regression model was used to test the association between the likelihood of having high-risk (Plaque-RADS 3 and 4) versus low-risk (Plaque-RADS 1 and 2) plaque subtypes and carotid plaque laterality. The model adjusted for age and sex and incorporated random intercepts for subjects to account for between-subject variability. Interaction terms between fixed effects were initially assessed using a factorial design; however, they were excluded from the final model due to a lack of statistical significance ($P>0.78$). Sensitivity analyses were performed using regressions after reclassifying plaques based on various CTA-based IPH volume thresholds, including >30 mm³, >40 mm³, and without any IPH-based classification. Statistical significance was prespecified at 0.05. Statistical analyses were performed using JMP, v 17.0 (SAS Institute, Inc, Cary, NC).

RESULTS

From 772 screened patients, 94 ($n=188$ carotid arteries) met inclusion criteria (Figure S1). Table S2 reports baseline characteristics. The Table reports the distribution of CTA-based Plaque-RADS subtypes. A total of 42.6% of plaques ipsilateral to stroke side were categorized as high risk, compared with 31.9% of plaques contralateral to the stroke side. There were no significant differences in baseline demographics across ipsilateral Plaque-RADS subtypes (Table S2).

Carotid plaques ipsilateral to the stroke side were significantly associated with higher-risk CTA-based Plaque-RADS subtypes (adjusted odds ratio, 2.10 [95% CI, 1.20–3.71]; $P=0.01$) compared with contralateral plaques, adjusting for age and sex. In sensitivity analyses, the significant association persisted, irrespective of IPH volume thresholds ($P<0.005$ for 30 and 40 mm³) or when excluding an IPH-based classification ($P=0.01$; Table S3).

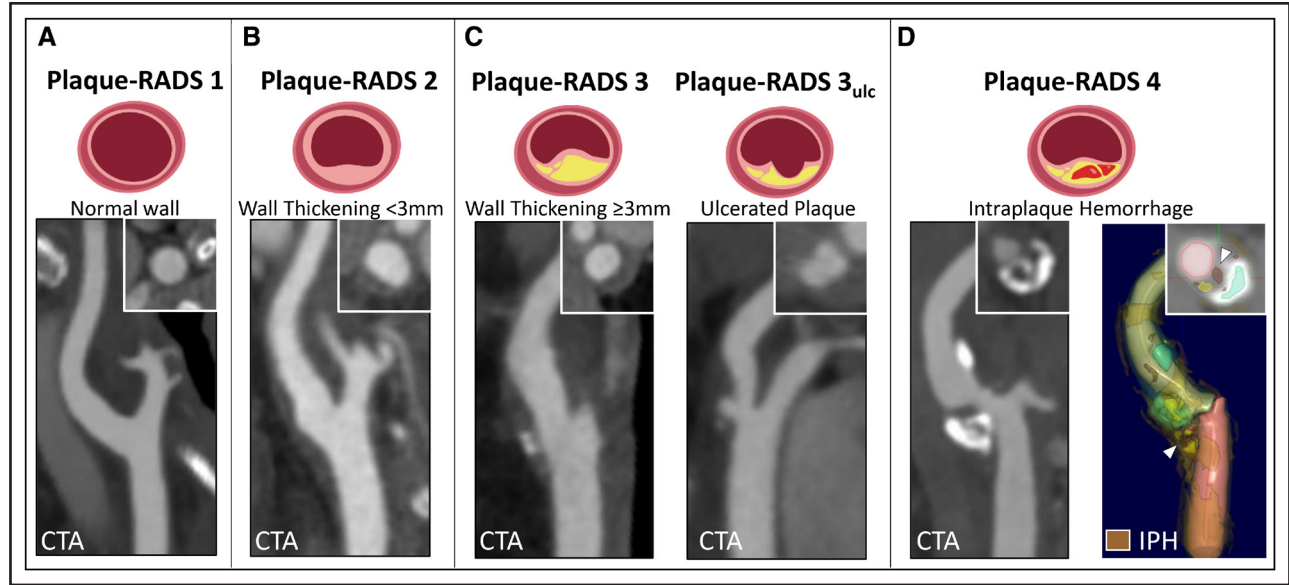


Figure. Computed tomography angiography (CTA)–based Plaque Reporting and Data System (Plaque-RADS) classification. CTA neck examples of the modified Plaque-RADS: (A) 1, (B) 2, (C) 3, and (D) 4. Plaque-RADS 4 shows the CTA and corresponding semiautomated intraplaque hemorrhage (IPH; arrowhead) segmentation. Plaque-RADS 3_{ulc} indicates Plaque-RADS 3_{ulceration}.

DISCUSSION

We show a significant association between high-risk CTA-based Plaque-RADS subtypes and the ipsilateral stroke side, suggesting this modified classification can be used to identify vulnerable plaques in patients with <50% carotid stenosis. Two prior studies have used Plaque-RADS to assess stroke risk related to carotid plaque. In a cohort of 61 patients with ≥50% carotid stenosis imaged by MR imaging, a higher proportion of high-risk Plaque-RADS subtypes in symptomatic versus asymptomatic carotid arteries was reported ($P<0.001$).⁶ However, generalizability was limited given the reliance on MR imaging. In a population-based retrospective cohort of 1378 patients with <70% stenosis, stratification with ultrasound/CT/MR-based Plaque-RADS <3 versus ≥3 predicted disease-free versus recurrence-free survival rates ($P<0.001$).⁷

Our study applies CTA-based Plaque-RADS to a well-defined ESUS cohort with <50% stenosis. A distinguishing criterion between low- versus high-risk Plaque-RADS subtypes is the 3 mm maximum wall thickness. This threshold is in keeping with studies of

mild-to-moderate stenotic plaque that show a ≥3-mm wall thickness threshold is more prevalent in carotid arteries ipsilateral to the stroke side.^{3,8} This criterion likely strengthens the applicability of this classification to mild-to-moderately stenotic plaques. However, further investigations are needed to understand how other features defined in Plaque-RADS can be applied to mild-to-moderately stenotic plaque. For example, conflicting results have been reported about ulcerations (Plaque-RADS 3C). In the prospective PARISK study (Plaque at RISK), ulcerated plaques with <70% stenosis showed no significant association with recurrent ipsilateral stroke at a mean follow-up of 5 years.⁹ By contrast, a retrospective cohort of 152 patients with ESUS identified plaque ulceration to be more prevalent in carotid arteries ipsilateral to the stroke side.¹⁰ Detecting ulcerations is challenging, as sensitivity may rely on technique (slice thickness) or mimic punctate intimal calcifications. Further study is needed to refine our understanding of plaque features that can be detected and classified accurately by CTA for mildly stenotic plaque.

Several limitations are noted. First, the contribution of cases with plaque ulceration and IPH to this data set was

Table. Distribution of Plaques by Plaque-RADS

Plaque-RADS subtypes, n (%)	All plaques (n=188)	Ipsilateral to stroke (n=94)	Contralateral to stroke (n=94)
1	34 (18.1)	14 (14.9)	20 (21.3)
2	84 (44.7)	40 (42.6)	44 (46.8)
3	69 (36.7)	39 (41.5)	30 (31.9)
3 _{ulceration}	3 (1.6)	1 (1.1)	2 (2.1)
4	1 (0.5)	1 (1.1)	0 (0)

Plaque-RADS indicates Plaque Reporting and Data System.

small. Future directions include retrospectively expanding this data set and also prospectively imaging patients using ultra-high spatial resolution photon-counting CTA to detect fibrous cap status and ulcerations/fissures.¹¹ Second, validation of this CTA-based Plaque-RADS system highlights the need for a prospective study to evaluate outcomes. Third, interobserver agreement across the different ultrasound/CT/MR modalities was previously assessed during the development of the Carotid Plaque-RADS system and reported a κ of 0.866.¹ However, interobserver agreement for our proposed modified CTA-based Plaque-RADS has not been investigated yet. Methods to assess and the reproducibility of categorizing Plaque-RADS 4 subcategories based on CT-attenuation characteristics of <25 Hounsfield units or the presence of the rim sign, both features associated with IPH, are areas of investigation.^{5,12} Fourth, in our study, a commercialized software was used to measure IPH volumes, a resource that may not be widely available. When limiting the subtypes to 1 and 2 for low risk and 3 only for high risk, plaques ipsilateral to the stroke side still had a higher prevalence of high-risk Plaque-RADS 3. The original Plaque-RADS system incorporates only the presence/absence of IPH rather than volumetric thresholds. Future directions include investigating the association of IPH volumes and plaque vulnerability. Given that clinical workflow and resource limitations may exist to combine multimodal ultrasound/CT/MR imaging features for plaque subtyping, a CTA-based system may facilitate the implementation of Plaque-RADS across centers.

In conclusion, we report a modified CTA-based Plaque-RADS classification for identifying vulnerable plaque in a cohort of patients with anterior circulation ESUS. Plaque-RADS could be an impactful tool in clinical practice and allow for risk stratification for vulnerable plaque phenotypes in ESUS.

ARTICLE INFORMATION

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Affiliations

Departments of Radiology (J.W.S., Y.S., J.H.W.), Biostatistics (Q.C., J.R.), and Neurology (L.I., S.E.Z., S.E.K., B.L.C.), University of Pennsylvania, PA. Drexel University College of Medicine, Philadelphia, PA (H.Q.P.). Cooper Medical School of Rowan University, Camden, NJ (M.K.). Department of Radiology, University of Cagliari, Italy (L.S.).

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Disclosures

None.

Supplemental Material

Supplemental Materials and Methods

Tables S1–S3

Figure S1

STROBE Checklist

REFERENCES

- Saba L, Cau R, Murgia A, Nicolaidis AN, Wintermark M, Castillo M, Staub D, Kakkos SK, Yang Q, Paraskevas KI, et al. Carotid plaque-RADS. *JACC Cardiovasc Imaging*. 2024;17:62–75. doi: 10.1016/j.jcmg.2023.09.005
- von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. The Strengthening of Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *J Clin Epidemiol*. 2008;61:344–349.
- Siegler JE, Thon JM, Woo JH, Do D, Messé SR, Cucchiara B. Prevalence of non-stenotic carotid plaque in stroke due to atrial fibrillation compared to embolic stroke of undetermined source. *J Stroke Cerebrovasc Dis*. 2019;28:104289. doi: 10.1016/j.jstrokecerebrovasdis.2019.07.005
- Sheahan M, Ma X, Paik D, Obuchowski NA, St Pierre S, Newman WP, Rae G, Perlman ES, Rosol M, Keith JC, et al. Atherosclerotic plaque tissue: noninvasive quantitative assessment of characteristics with software-aided measurements from conventional CT angiography. *Radiology*. 2018;286:622–631. doi: 10.1148/radiol.2017170127
- Saba L, Micheletti G, Brinjikji W, Garofalo P, Montisci R, Balestrieri A, Suri JS, DeMarco JK, Lanzino G, Sanfilippo R. Carotid intraplaque-hemorrhage volume and its association with cerebrovascular events. *AJNR Am J Neuroradiol*. 2019;40:1731–1737. doi: 10.3174/ajnr.A6189
- Xia J, Yu C, Li L, Pan J. Arterial transit artifacts and carotid plaque-RADS may predict symptoms in patients with carotid stenosis. *Magn Reson Imaging*. 2024;111:131–137. doi: 10.1016/j.mri.2024.05.001
- Huang Z, Cheng XQ, Lu RR, Bi XJ, Liu YN, Deng YB. Incremental prognostic value of carotid plaque-RADS over stenosis degree in relation to stroke risk. *JACC Cardiovasc Imaging*. 2025;18:77–89. doi: 10.1016/j.jcmg.2024.07.004
- Coutinho JM, Derkatch S, Potvin ARJ, Tomlinson G, Kiehl TR, Silver FL, Mandell DM. Nonstenotic carotid plaque on CT angiography in patients with cryptogenic stroke. *Neurology*. 2016;87:665–672. doi: 10.1212/WNL.0000000000002978
- van Dam-Nolen DHK, Truijman MTB, van der Kolk AG, Liem MI, SchreuderFHB, Boersma E, Daemen MJAP, Mess WH, van Oostenbrugger RJ, van der Steen AFW, et al. PARISK Study Group. Carotid plaque characteristics predict recurrent ischemic stroke and TIA: the PARISK (Plaque at RISK) study. *JACC Cardiovasc Imaging*. 2022;15:1715–1726. doi: 10.1016/j.jcmg.2022.04.003
- Jumah A, Aboul Nour H, Intikhab O, Choudhury O, Gagi K, Fana M, Alhajala H, Alkhouljah M, Alsrouji OK, Eltous L, et al. Non-stenosing carotid artery plaques in embolic stroke of undetermined source: a retrospective analysis. *Neurol Sci*. 2023;44:247–252. doi: 10.1007/s10072-022-06425-w
- Shami A, Sun J, Gialeli C, Markstad H, Edsfieldt A, Aurumskjöld M-L, Gonçalves I. Atherosclerotic plaque features relevant to rupture-risk detected by clinical photon-counting CT ex vivo: a proof-of-concept study. *Eur Radiol Exp*. 2024;8:14. doi: 10.1186/s41747-023-00410-4
- Benson JC, Nardi V, Madhavan AA, Bois MC, Saba L, Savastano L, Lerman A, Lanzino G. Reassessing the carotid artery plaque “Rim Sign” on CTA: a new analysis with histopathologic confirmation. *AJNR Am J Neuroradiol*. 2022;43:429–434. doi: 10.3174/ajnr.A7443